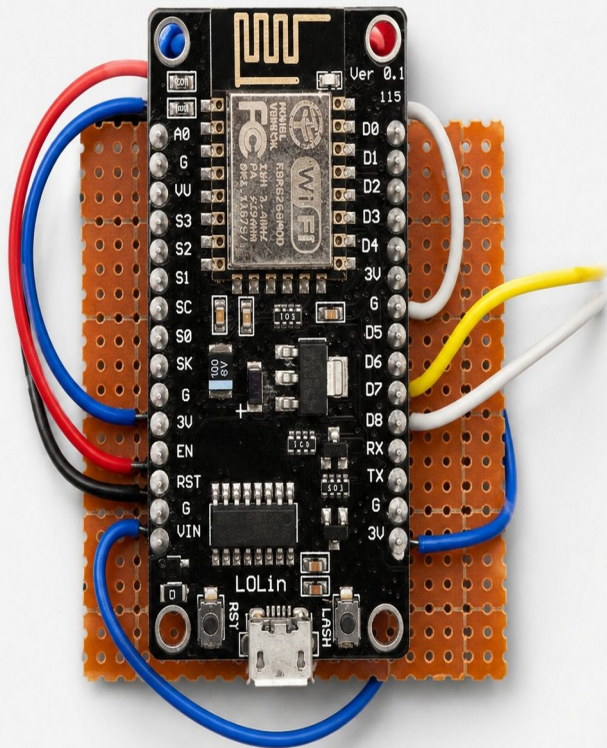




IoT Course Project

Smart Room Automation System

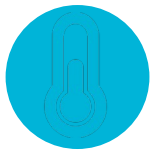
Automated lighting and fan control
using PIR and DHT sensors
via Blynk IoT platform

ESP8266 NodeMCU • Blynk Cloud • C++ Arduino

WHAT IS MY PROJECT?

Smart Room Automation System

An IoT system that automatically controls a room's fan and lighting based on real-time sensor data, and allows remote monitoring and manual override via a website or smartphone app.



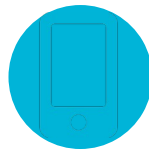
Senses the Environment

DHT22 measures temperature & humidity every 2 seconds. If temperature exceeds 26°C, the fan turns ON automatically.



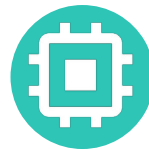
Detects Motion

PIR sensor detects if someone is in the room. When motion is detected, the LED light turns ON instantly.



Remote Control

All data is sent to the Blynk app via Wi-Fi. You can monitor temperature, humidity, fan and light status from your phone.



Powered by ESP32

The ESP32 microcontroller runs all the logic, reads sensors, controls actuators, and maintains the Wi-Fi connection.

PROJECT OVERVIEW

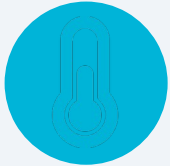
The Problem

Traditional rooms waste energy because lights and fans run even when no one is present.

There is no automatic response to temperature changes or occupancy.

Our Solution

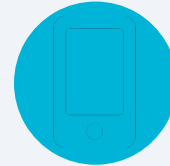
An ESP8266-based IoT system that automatically turns the fan ON when temperature is high, and turns the LED light ON when motion is detected — all monitored via the Blynk app.



**Temperature
& Humidity**

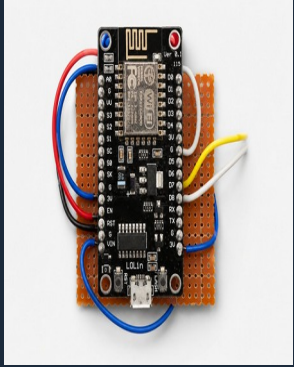


**Motion-Based
Lighting**



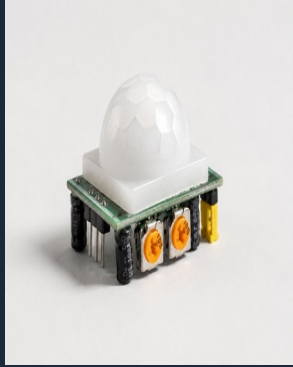
**Remote
Monitoring**

COMPONENTS USED



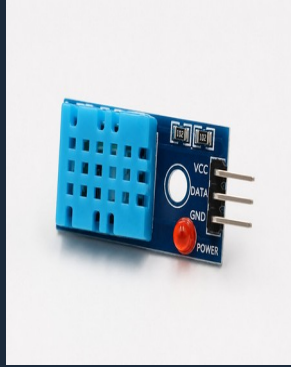
**ESP8266
NodeMCU**

Main controller
with built-in Wi-Fi



**PIR
Sensor**

Detects motion
in the room



**DHT11
Sensor**

Measures temperature
& humidity



**Relay
Module**

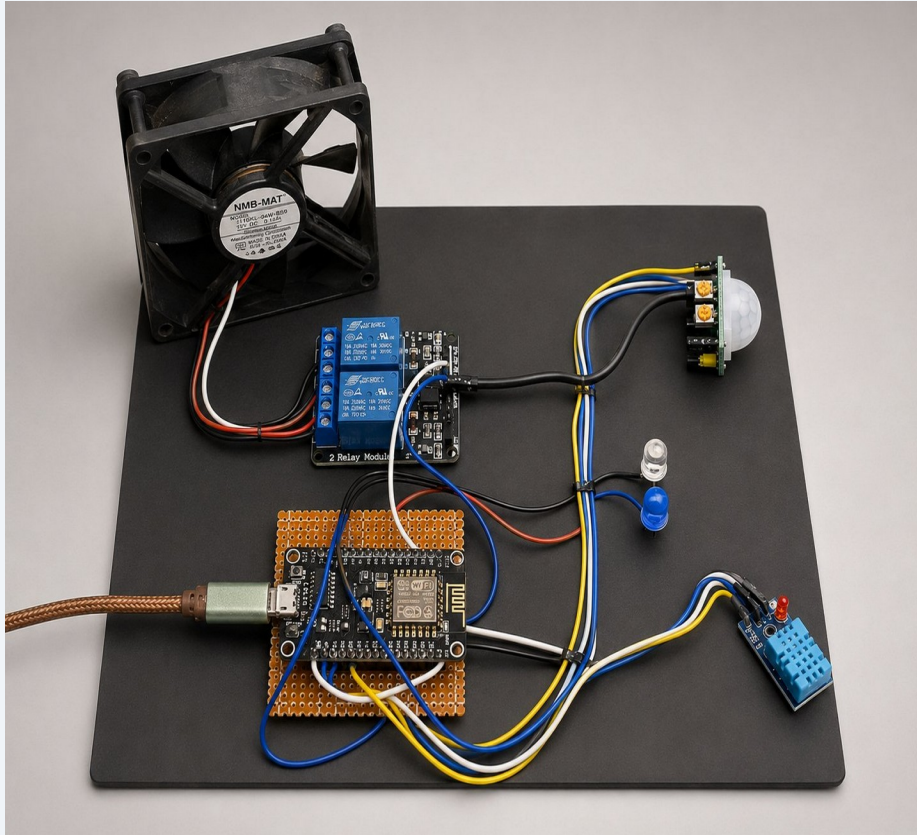
Switches fan
power on/off



**LED
Light**

Represents
room lighting

SYSTEM ARCHITECTURE



Sensors Read Data

PIR detects motion
DHT reads temp & humidity

ESP8266 Processes

Logic checks thresholds
and motion status

Actuators Respond

Relay → Fan ON/OFF
LED → Light ON/OFF

Blynk Dashboard

Real-time monitoring
Manual override available

HOW IT WORKS



Light Control

IF motion is detected (PIR = HIGH)

→ **Turn LED ON** (`lightStatus = 1`)

OR if Manual Light toggle is ON

→ **Turn LED ON**

ELSE → LED OFF



Fan Control

IF temperature > 26°C (threshold)

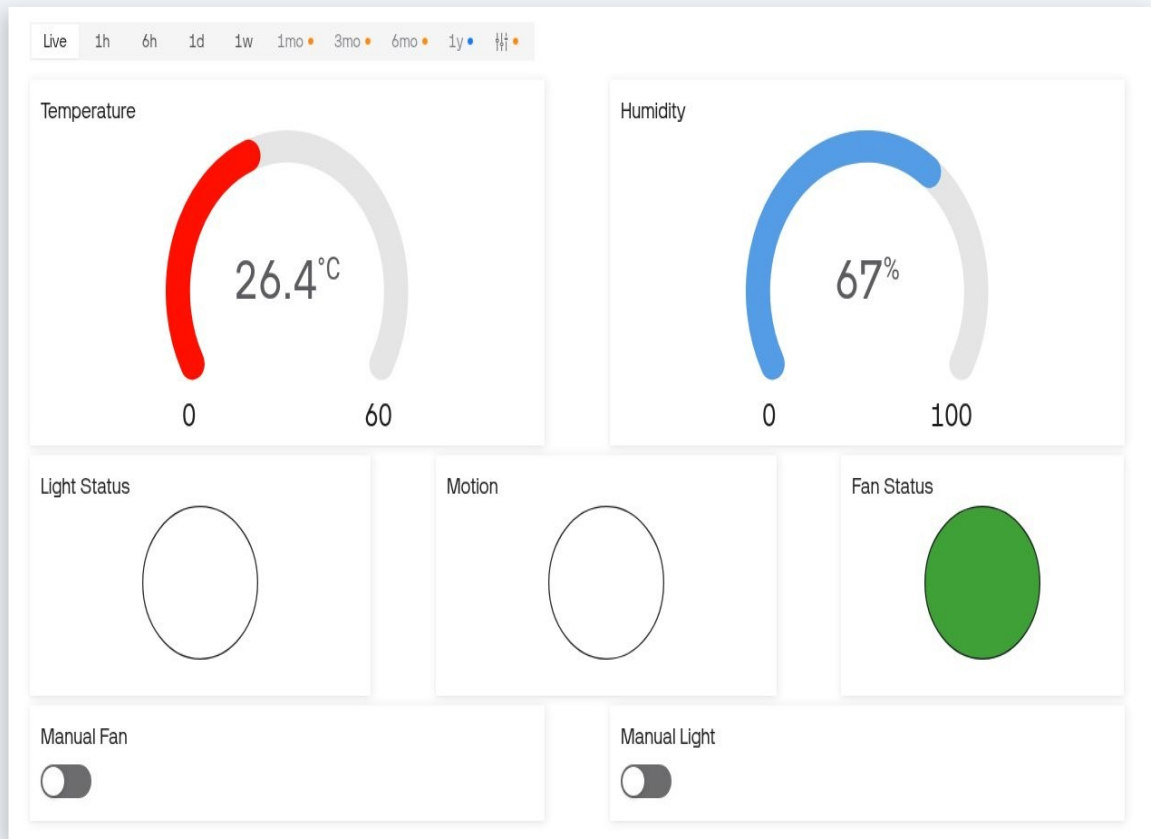
→ **Turn Fan ON via Relay** (`fanStatus = 1`)

OR if Manual Fan toggle is ON

→ **Turn Fan ON**

ELSE → Fan OFF

BLYNK DASHBOARD



V0 — Temperature gauge (°C)



V1 — Humidity gauge (%)



V4 — Light Status indicator



V2 — Motion indicator

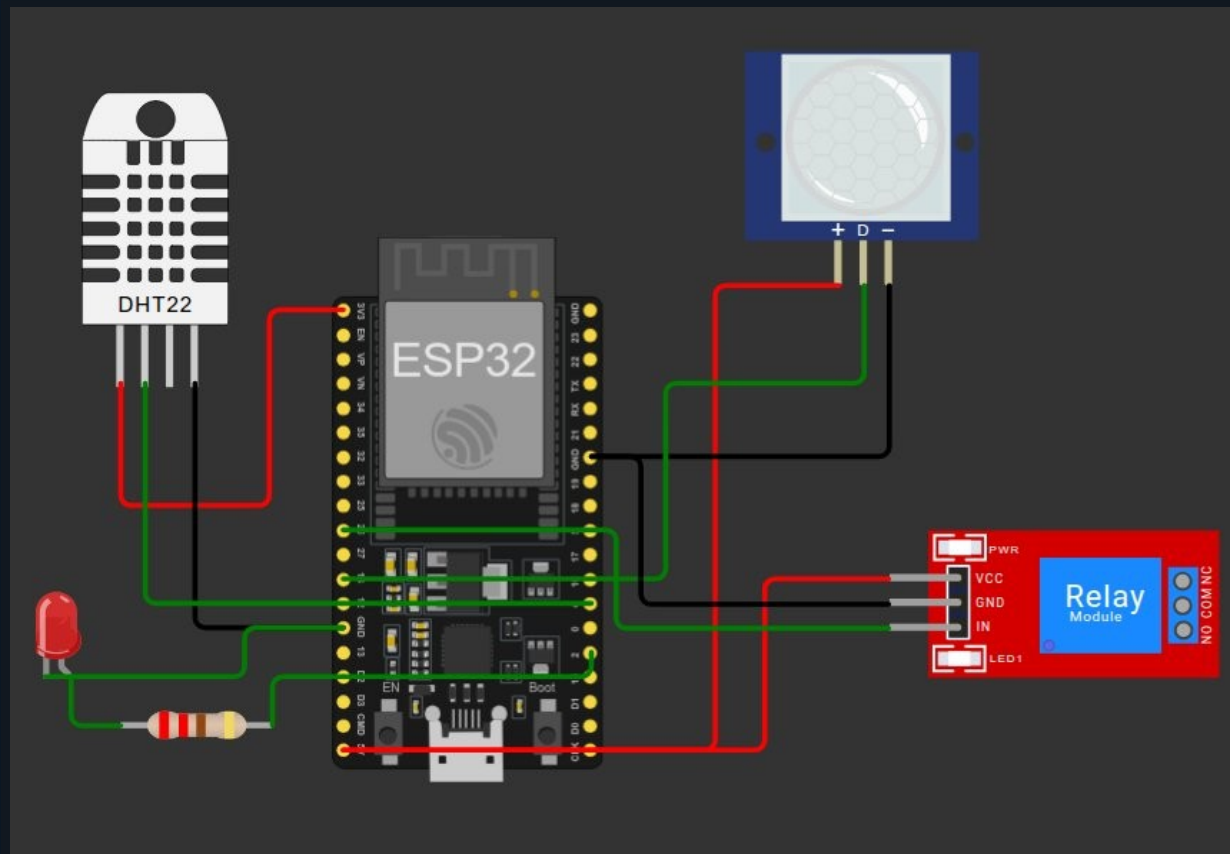


V3 — Fan Status indicator



V5/V6 — Manual toggles

CIRCUIT WIRING DIAGRAM



RED

3.3V / VCC Power



GREEN

Data / Signal lines



BLACK

GND (Ground)

Pin Connections

DHT22 → GPIO 4

PIR → GPIO 14

LED → GPIO 2

Relay → GPIO 26

THE CODE

```
1  #define BLYNK_TEMPLATE_ID "TMPL4ej1EWACs"
2  #define BLYNK_TEMPLATE_NAME "Smart Home System"
3  #define BLYNK_AUTH_TOKEN "6-H4mmWMzrInpak9vjDxHezvSdFYmt6-"
4
5  #include <ESP8266WiFi.h>
6  #include <BlynkSimpleEsp8266.h>
7  #include <DHT.h>
8
9  char ssid[] = "EMRE DEVREM";
10 char pass[] = "Marketing@2999";
11
12 #define DHTPIN D4
13 #define DHTTYPE DHT11
14
15 #define PIR_PIN D6
16 #define LED_PIN D1
17 #define RELAY_PIN D8
18
19 DHT dht(DHTPIN, DHTTYPE);
20 BlynkTimer timer;
21
22 float tempThreshold = 26.0;
23
24 int manualFan = 0;
25 int manualLight = 0;
26
27 BLYNK_WRITE(V5)
28 {
29   manualFan = param.asInt();
30 }
31
32 BLYNK_WRITE(V6)
33 {
34   manualLight = param.asInt();
35 }
36
37 void sendSensorData()
38 {
39   float temp = dht.readTemperature();
40   float hum = dht.readHumidity();
41   int motion = digitalRead(PIR_PIN);
42
43   if (isnan(temp) || isnan(hum)) {
44     Serial.println("DHT11 reading error");
45     return;
46   }
47 }
```

```
47
48 int lightStatus = 0;
49 int fanStatus = 0;
50
51 // Light control
52 if (motion == HIGH || manualLight == 1) {
53   digitalWrite(LED_PIN, HIGH);
54   lightStatus = 1;
55 } else {
56   digitalWrite(LED_PIN, LOW);
57   lightStatus = 0;
58 }
59
60 // Fan control
61 if (temp > tempThreshold || manualFan == 1) {
62   digitalWrite(RELAY_PIN, HIGH);
63   fanStatus = 1;
64 } else {
65   digitalWrite(RELAY_PIN, LOW);
66   fanStatus = 0;
67 }
68
69 // Send data to Blynk
70 Blynk.virtualWrite(V0, temp);
71 Blynk.virtualWrite(V1, hum);
72 Blynk.virtualWrite(V2, motion);
73 Blynk.virtualWrite(V3, fanStatus);
74 Blynk.virtualWrite(V4, lightStatus);
75
76 Serial.print("Temperature: ");
77 Serial.print(temp);
78 Serial.print(" C | Humidity: ");
79 Serial.print(hum);
80 Serial.print(" % | Motion: ");
81 Serial.print(motion);
82 Serial.print(" | Fan: ");
83 Serial.print(fanStatus);
84 Serial.print(" | Light: ");
85 Serial.println(lightStatus);
86
87 }
```

```
87
88 void setup()
89 {
90   Serial.begin(115200);
91
92   pinMode(PIR_PIN, INPUT);
93   pinMode(LED_PIN, OUTPUT);
94   pinMode(RELAY_PIN, OUTPUT);
95
96   digitalWrite(LED_PIN, LOW);
97   digitalWrite(RELAY_PIN, LOW);
98
99   dht.begin();
100
101   Blynk.begin(BLYNK_AUTH_TOKEN, ssid, pass);
102
103   timer.setInterval(2000L, sendSensorData);
104 }
105
106 void loop()
107 {
108   Blynk.run();
109   timer.run();
110 }
111 }
```

RESULTS & TESTING



Motion Detection

✓ LED turned ON within 1 second of detecting presence



Temperature Control

✓ Fan activated when temp exceeded 26°C threshold (tested at 26.4°C)



DHT11 Reading

✓ Accurate readings: 26.4°C temperature, 67% humidity (live dashboard)



Manual Override

✓ Blynk toggle switches worked to control fan and light independently



Wi-Fi Connectivity

✓ Data refreshed every 2 seconds via Blynk cloud with no disconnections



Conclusion

- Successfully automated room lighting using PIR motion detection
- Fan automation based on real-time temperature threshold
- Remote monitoring and manual control via Blynk mobile app
- Low-cost, scalable IoT solution using ESP8266

Future Improvements

- Add sound sensor for noise-based automation
- Integrate with Google Assistant / Alexa
- Add energy usage tracking and analytics